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PredictRisk: A Professional Bayesian Web Application for Cardiovascular Risk Assessment

Taiwo Michael Ayeni [1]

1 Northeastern University, College of Professional Studies, Analytics, Toronto, Canada

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Software

- Review
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Summary

PredictRisk is a comprehensive Python package for Bayesian cardiovascular risk assessment with explicit uncertainty quantification. The software implements a modular architecture with eight specialized modules, demonstrating substantial scholarly effort in both statistical methodology and software engineering. Unlike existing cardiovascular risk tools that provide only point estimates, PredictRisk uniquely combines advanced Bayesian inference with comprehensive uncertainty quantification, professional software engineering practices, and a complete synthetic data generation framework. The package features six independent Bayesian logistic regression models for major cardiovascular conditions, a sophisticated web interface built with Streamlit, comprehensive PDF report generation, and extensive validation frameworks. PredictRisk has been successfully deployed and is actively used for cardiovascular risk research and education, distinguishing itself through its focus on uncertainty communication, open-source methodology, and modular architecture designed for extensibility.

Statement of need

Cardiovascular diseases remain the leading cause of global mortality, yet existing risk assessment tools suffer from significant limitations (Roth et al., 2020). Current clinical tools like the Framingham Risk Score and ASCVD Risk Calculator provide only point estimates without uncertainty quantification, limiting their utility for informed decision-making (D'Agostino Sr et al., 2008; Goff Jr et al., 2014). Unlike existing health informatics packages that focus on specific aspects of medical data analysis, PredictRisk uniquely addresses the critical gap of uncertainty-aware cardiovascular risk assessment.

Unique Contributions Compared to Existing Software:

While the health informatics landscape includes packages for electrocardiogram processing, survival analysis, and epidemiological modeling, PredictRisk is specifically designed for comprehensive cardiovascular risk assessment with several distinctive features:

1. **Bayesian Uncertainty Quantification**: Unlike deterministic risk calculators, PredictRisk provides full posterior distributions and credible intervals for every prediction
2. **Comprehensive Synthetic Data Framework**: Includes complete data generation methodology with realistic correlations, enabling privacy-preserving research and education
3. **Multi-Condition Assessment**: Simultaneous modeling of six major cardiovascular conditions with consistent methodology
4. **Educational Focus**: Designed specifically for teaching Bayesian methods in healthcare applications with transparent methodology
5. **Professional Package Architecture**: Modular design supporting easy extension and integration into larger healthcare systems

From a software engineering perspective, most existing cardiovascular risk tools are proprietary,

closed-source systems that cannot be modified, extended, or validated by the research community. This creates barriers to methodological improvements, limits educational applications, and prevents researchers from building upon existing work.

PredictRisk fills this gap by providing an open-source, professionally-engineered platform that combines sophisticated Bayesian statistical methods with user-friendly interfaces. The software serves multiple user communities: (1) researchers studying cardiovascular risk modeling methodologies, (2) healthcare professionals seeking accessible risk assessment tools with uncertainty quantification, (3) data scientists learning Bayesian approaches to medical prediction, (4) educators teaching statistical methods in healthcare, and (5) individuals interested in understanding their cardiovascular risk profile.

Comparison with Existing Software

While several health informatics packages exist in the Python ecosystem, PredictRisk addresses a specific gap in cardiovascular risk assessment:

- **Pyheartlib**: Focuses on electrocardiogram signal processing, while PredictRisk provides risk prediction and assessment
- **lifelines**: Provides survival analysis tools, while PredictRisk specializes in Bayesian cardiovascular risk modeling
- **epimargin**: Offers epidemiological estimation tools, while PredictRisk focuses specifically on individual cardiovascular risk assessment with uncertainty quantification
- **General medical packages**: Most existing tools provide deterministic predictions without uncertainty quantification

PredictRisk's unique combination of Bayesian uncertainty quantification, multi-condition assessment, synthetic data generation, and educational focus distinguishes it from existing health informatics software.

Software description

Architecture and Design

PredictRisk implements a sophisticated modular architecture with eight specialized Python modules, following established software engineering best practices with proper separation of concerns, comprehensive error handling, and extensive documentation throughout.

Core Architecture Components:

- **Configuration Management** (`predictrisk.config`): Centralized configuration with input validation ranges, model parameters, and application settings
- **Risk Assessment Engine** (`predictrisk.risk_calculator`): Core Bayesian risk assessment with model loading, prediction, batch processing, and convergence validation
- **Utility Framework** (`predictrisk.utils`): Comprehensive utility functions for data validation, statistical calculations, and error handling
- **Recommendation System** (`predictrisk.advice_engine`): Sophisticated advice generation with personalized recommendations and educational content
- **Report Generation** (`predictrisk.pdf_generator`): Professional PDF report creation with multiple formats and comprehensive content organization
- **Data Generation Framework** (`predictrisk.data_generator`): Advanced synthetic data generation with realistic correlations and validation capabilities
- **Web Interface** (`predictrisk.web_app`): Modular Streamlit interface with professional UI/UX design
- **Package Infrastructure**: Professional setup.py, comprehensive imports, and command-line entry points

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Bayesian Statistical Framework

The statistical foundation employs six independent Bayesian logistic regression models implemented through the Bambi library (Capretto et al., 2022), providing a high-level interface to PyMC (Salvatier et al., 2016). Each model demonstrates exceptional convergence properties (R-hat = 1.0) with effective sample sizes exceeding 2,570 for all parameters, utilizing No-U-Turn Sampler (NUTS) with four parallel chains and 1,000 draws per chain after 1,000 tuning steps.

Distinctive Technical Features:

Unlike similar health informatics packages, PredictRisk provides unique technical capabilities: - **Multi-condition Bayesian Framework**: Simultaneous assessment of six cardiovascular conditions using consistent methodology - **Uncertainty Quantification Focus**: Every prediction includes full posterior distributions and credible intervals - **Synthetic Data Integration**: Complete data generation framework with realistic correlations for educational and research applications - **Modular Architecture**: Professional package structure enabling easy extension and integration - **Research-Education Bridge**: Explicitly designed for both practical applications and methodological education

Overview of Features

- **Multi-Condition Risk Assessment**: PredictRisk can simultaneously assess risk for six major cardiovascular conditions using consistent Bayesian methodology. Users can easily extend the framework to include additional conditions by following the established model structure.

- **Comprehensive Uncertainty Quantification**: Unlike deterministic risk calculators, every prediction includes full posterior distributions, 95% credible intervals, and detailed uncertainty analysis. The framework provides multiple confidence levels and statistical summaries.

- **Professional Web Interface**: The Streamlit-based interface provides an intuitive user experience with real-time validation, educational tooltips, and responsive design. The interface supports both individual assessments and educational demonstrations.

- **Advanced Report Generation**: PredictRisk automatically generates comprehensive PDF reports with professional formatting, risk visualization, personalized recommendations, and educational content. Reports can be customized for different use cases.

- **Synthetic Data Framework**: The package includes a complete synthetic data generation system with realistic correlations, enabling privacy-preserving research and educational applications. Data validation and quality assessment tools are built-in.

- **Extensible Architecture**: The modular design allows researchers to easily add new cardiovascular conditions, risk factors, or analysis methods. The package supports both programmatic usage and web-based interaction.

- **Educational Integration**: PredictRisk is specifically designed for teaching Bayesian methods in healthcare applications, with comprehensive documentation, example notebooks, and transparent methodology.

- **Production Deployment**: The software is successfully deployed at https://predictrisk.streamlit.app/ and has been used for cardiovascular risk research and educational applications, demonstrating real-world utility and reliability.

Quality assurance

Software Engineering Standards

PredictRisk implements professional software development practices:

- **Package Structure**: Proper Python package organization with setup.py for pip installation

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- **Code Quality**: Comprehensive docstrings, type hints, and inline documentation throughout
- **Error Handling**: Robust exception handling with detailed logging and user-friendly error messages
- **Input Validation**: Extensive validation with range checking, type conversion, and error reporting
- **Modular Architecture**: Clean separation of concerns enabling independent module development and testing
- **Performance Optimization**: Efficient model loading, caching, and responsive user interface design

Statistical and Model Validation

The Bayesian models undergo rigorous validation:

- **Convergence Assessment**: All models achieve R-hat = 1.0 with effective sample sizes > 2,570
- **Cross-validation**: Model performance validation using holdout datasets
- **Posterior Predictive Checks**: Model adequacy assessment through posterior predictions
- **Synthetic Data Validation**: Comprehensive quality checking of generated datasets
- **Reproducibility**: Complete seed-based reproducibility for all statistical operations

Deployment and Testing

- **Live Deployment**: Functional web application deployed at https://predictrisk.streamlit.app/
- **Local Testing**: Package successfully installs via pip and runs locally
- **Integration Testing**: All modules work together seamlessly in the web application
- **User Interface Testing**: Comprehensive input validation and error handling verification
- **Cross-platform Compatibility**: Tested on Windows, Mac, and Linux environments

Community guidelines

Open Source Development

PredictRisk embraces professional open-source development practices. The project is hosted on GitHub (https://github.com/ayeni-T/PredictRisk) under the MIT License, ensuring broad accessibility and enabling community contributions. The modular architecture facilitates collaborative development with clear module boundaries and comprehensive documentation.

Contributing Framework

The package provides multiple pathways for community engagement:

- **Bug Reports and Feature Requests**: GitHub Issues system for community feedback
- **Code Contributions**: Pull request workflow for improvements, new features, and bug fixes
- **Clinical Validation**: Collaboration framework for validation studies using real clinical datasets
- **Model Extensions**: Architecture supports the addition of new cardiovascular conditions and risk factors
- **Documentation Enhancement**: Community contributions to user guides, tutorials, and API documentation
- **Educational Applications**: Framework for developing teaching materials and case studies

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Development and Extension

The modular architecture enables several enhancement pathways:

1. **Clinical Integration**: FHIR-compliant APIs for electronic health record integration
2. **Advanced Analytics**: SHAP/LIME integration for model interpretability and feature importance
3. **Longitudinal Modeling**: Time-to-event analysis and dynamic risk assessment capabilities
4. **Multi-language Support**: Internationalization framework for global accessibility
5. **Mobile Optimization**: Responsive design improvements for mobile device compatibility
6. **Advanced Visualization**: Interactive plotting and risk visualization enhancements

Technical implementation

Installation and Usage

PredictRisk can be easily installed via pip from the GitHub repository:

```
pip install git+https://github.com/ayeni-T/PredictRisk.git
```

Web Application Usage:

```
# Launch the web interface
streamlit run app.py

# Or use the modular interface
python -m predictrisk.web_app
```

Programmatic Usage:

```
from predictrisk import RiskCalculator, generate_synthetic_data

# Create risk calculator
calculator = RiskCalculator()

# Assess cardiovascular risk
input_data = {
    "age": 55, "bmi": 28.5, "glucose": 110,
    "systolic_bp": 145, "smoking_status": 1,
    # ... other risk factors
}

risk, credible_interval, summary = calculator.predict_risk("Stroke", input_data)
print(f"Stroke risk: {risk:.2%} [{credible_interval[0]:.2%}, {credible_interval[1]:.2%}]

# Generate synthetic data for research
synthetic_data = generate_synthetic_data(n_samples=1000, random_seed=42)
```

Command-Line Tools:

```
# Generate synthetic dataset
predictrisk-generate-data --samples 1000 --output mydata.csv

# Launch web application
predictrisk-app
```

Performance and Scalability

- **Model Loading**: Efficient NetCDF4-based model persistence with fast loading times

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- **Prediction Speed**: Real-time risk assessment with sub-second response times
- **Memory Management**: Optimized memory usage for large-scale applications
- **Concurrent Users**: Web application supports multiple simultaneous users
- **Batch Processing**: Efficient batch prediction capabilities for research applications

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